

Supplementary Information

Resonant versus geometric anisotropy of the Purcell enhancement near a gold nanorod

S1. Boundary-element calculations: validation and convergence

All boundary-element (BEM) results were obtained with MNPBEM17 [3–5] (retarded solver, curved boundary elements, Johnson–Christy gold from the tabulated file distributed with the toolbox) in MATLAB R2026b on GitHub-hosted runners. The script `isi/bem/run_mnpbem.m` defines all cases; the workflow `.github/workflows/bem-runs.yml` runs one case per job. Table S1 compares the equal-volume sphere with the exact Mie solution evaluated with the same tabulated permittivity (radial dipole: Ruppel [6]; tangential dipole: Kim *et al.* [7]); Table S2 and Fig. S1 give the mesh convergence of the rod.

Table S1: BEM versus exact Mie theory for the equal-volume gold sphere ($R = 15.874$ nm). Median absolute relative deviation over 500 nm to 900 nm (maximum in brackets for the radial F_p).

mesh (vertices)	host	gap (nm)	radial F_p	radial T	tangential F_p	tangential T
400	air	5	2.4% (6.5%)	1.8%	2.1%	1.0%
784	air	5	1.0% (4.4%)	1.4%	0.7%	0.9%
1444	air	5	0.9% (4.2%)	1.4%	0.7%	1.0%
1444	air	10	1.5% (3.8%)	1.3%	0.5%	1.1%
1444	air	20	0.9% (3.3%)	1.3%	0.1%	1.2%
784	water	5	0.6% (4.4%)	1.4%	0.7%	0.8%
784	water	20	0.4% (2.9%)	1.3%	0.1%	1.2%

Table S2: Mesh convergence of the rod (axial dipole). h : target element size. ^aWavelengths 500 nm to 808 nm only (six-hour limit of the CI job).

h (nm)	elements	gap (nm)	$F_p^{\max}$ (λ)	$T_{\max}$ (λ)	$\eta_a(\lambda_T)$
3	720	5	1649 (604)	117.7 (608)	7.3%
2	1550	5	1643 (604)	117.0 (608)	7.3%
1.5	2829	5	1638 (604)	116.5 (608)	7.3%
1 ^a	6324	5	1634 (604)	116.2 (608)	7.3%
3	720	3	3672 (508)	209.8 (608)	6.5%
1.5	2829	3	3782 (508)	214.1 (608)	6.6%
1 ^a	6324	3	3770 (508)	213.1 (608)	6.6%
1.5	2829	20	78 (604)	8.2 (612)	11.6%

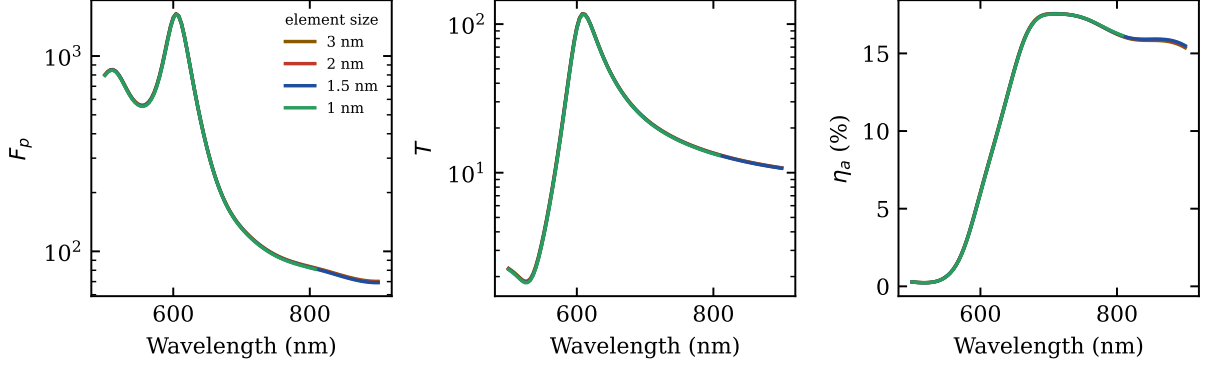


Figure S1: Mesh convergence of the BEM for the rod with axial dipole at a 5 nm gap.

S2. Plasmon-eigenmode decomposition, excitation and additional set-ups

Eigenmodes. The 40 lowest eigenvalues of the quasistatic BEM matrix F were computed with `bemstateig` of MNPBEM17 on the 2 nm meshes (sphere: 784 vertices). The longitudinal (transverse) dipolar mode is the eigenvector u_k with the largest oscillator strength $s_k = (\sum_j a_j \mathbf{r}_j u_{k,j}) \cdot (\tilde{u}_k \cdot \mathbf{n})$ for a uniform field along z (x), where a_j , $\mathbf{r}_j$ and $\mathbf{n}_j$ are the area, centroid and normal of element j ; its contribution to $F_p - 1$ follows from Eq. (3) of the main text and is evaluated with the same decay-rate routine as the full solution. The trivial eigenvalue -2π (uniform charge) does not couple. Script: `isi/bem/extra/job_modes.m`. Table S3 lists the decomposition at the resonance of the longitudinal mode.

Table S3: Quasistatic decomposition of the axial F_p at the resonance of the longitudinal dipolar mode (λ_L). Share: fraction of $F_p - 1$ carried by that mode; background: $F_p - 1$ minus that contribution, plus one; geometric ratio: background divided by the transverse F_p ; total ratio: axial over transverse F_p .

particle	λ_L (nm)	gap (nm)	axial F_p	share	background	geometric ratio	total ratio
rod AR 2	540	5	859	57%	368	2.7	6
		10	154	80%	32	2.6	13
rod AR 3	595	3	2936	80%	593	2.7	13
		5	1374	91%	129	3.1	33
		10	352	96%	14	3.3	82
		20	55	98%	1.9	1.6	47
rod AR 4	655	5	3095	98%	57	3.4	185
		10	861	99%	7.5	3.3	382
rod AR 5	725	5	4661	99%	46	3.7	374
		10	1322	100%	6.8	3.6	693
sphere	510	5	761	9%	693	2.4	3
		10	89	21%	70	2.4	3

Retarded pole fits. The full BEM spectra $F_p(\omega) - 1$ of the axial dipole were fitted over the window given below with $\text{Im}[C/(\tilde{\omega} - \omega)] + a_0 + a_1(\omega - \omega_0) + a_2(\omega - \omega_0)^2$ (C , $\tilde{\omega}$ complex; relative residuals), the quasinormal-mode form of a single modal contribution plus a smooth background. The method was first applied to the quasistatic spectra, for which the dipolar-mode term is known exactly (Table S3). Script: `isi/analysis/pole_fit.py`.

Shape of the rod ends. The caps of the $L = 60$ nm, $D = 20$ nm rod were replaced by half-

Table S4: Share of $F_p - 1$ carried by the longitudinal dipolar mode at its resonance from pole fits. Quasistatic: exact value / fitted value. Retarded: fitted value, pole wavelength and full linewidth.

aspect ratio	gap (nm)	quasistatic exact / fit	retarded fit	λ (nm)	width (meV)
2	5	57% / $-^a$	$-^a$	545	–
2.5	5	–	74%	582	237
3	3	80% / 59%	73%	614	145
3	5	91% / 78%	88%	610	133
3	10	96% / 89%	94%	608	132
3	20	98% / 94%	96%	607	133
4	5	98% / 97%	97%	676	86
4	10	99% / 98%	97%	676	89
5	5	99% / 98%	100%	756	85
5	10	100% / 99%	101%	756	85

^aThe resonance lies at the interband edge; the fit cannot separate mode and background (fitted share >100%).

spheroids with axial semi-axis $c = rR$ by an affine map of the hemispherical cap vertices and curved-element midpoints (same total length; `bemx_tiprod.m`, `job_tips.m`). Retarded BEM and quasistatic decomposition as above.

Table S5: Effect of the end-cap shape (axial dipole on the axis). λ_T , $T_{\max}$, η_a : radiative peak; background ratio and mode share: quasistatic decomposition at the resonance of the dipolar mode.

$r = c/R$	gap (nm)	λ_T (nm)	$T_{\max}$	η_a	$\max F_p^{\parallel}/F_p^{\perp}$	background ratio
1	5	610	117.5	7.6%	49	3.1
1	10	610	36.3	8.5%	135	3.3
0.75	5	615	128.9	8.2%	47	2.6
0.75	10	615	42.5	9.0%	135	2.6
0.5	5	620	144.6	8.8%	45	2.1
0.5	10	620	51.9	9.4%	135	2.0
0.35	5	625	157.1	9.5%	43	1.8
0.35	10	625	60.6	10.1%	133	1.7

Substrate. The rod (axis along x) lies 1 nm above a glass half-space ($\varepsilon = 1.52^2$); the reflected Green function is tabulated with `compgreentablayer` at 23 wavelengths between 480 and 920 nm and interpolated. Rates are given in units of the vacuum rate; the reference is the same dipole above bare glass (`decayrate0`). Script: `job_substrate.m`.

Collection on glass by reciprocity. The angular emission of the dipole next to the rod on glass was obtained from $dP/d\Omega = n_j \sum_{\text{pol}} |\mathbf{p} \cdot \mathbf{E}_{\text{loc}}(\hat{\mathbf{s}}, \text{pol})|^2$, where $\mathbf{E}_{\text{loc}}$ is the field at the dipole for a unit-amplitude plane wave incident from direction $\hat{\mathbf{s}}$ in medium j (air or glass), computed with the layer BEM for 24×36 directions per half-space and two polarisations (3456 right-hand sides of one BEM solution; `job_recip.m`). A free dipole gives $8\pi/3$, which fixes the absolute normalisation. The formula was verified against an independent angular-spectrum (Fresnel) calculation for a bare dipole above glass (`isi/analysis/glass_check.py`). The far field returned directly by the layer routines of MNPBEM17 was not used: the induced-charge and dipole contributions are returned on differently ordered direction sets, so that their sum, and quantities derived from it, are inconsistent for weakly radiating dipoles. Table S6 lists the checks and results.

Nonlocal response. Following Luo *et al.* [8] the metal is shrunk by $\Delta d = 0.05$ nm and covered by a layer of the same thickness with permittivity $\varepsilon_t = \varepsilon_m \varepsilon_b q_L \Delta d / (\varepsilon_m - \varepsilon_b)$, $q_L =$

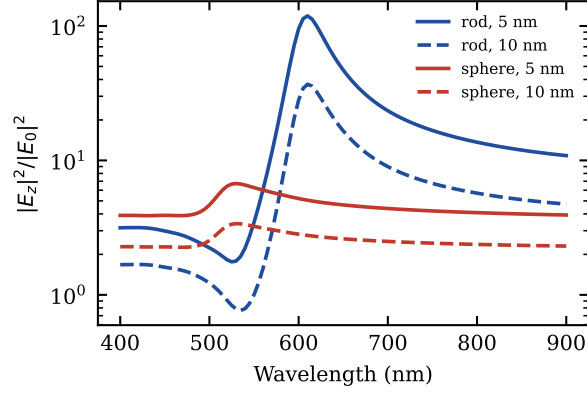


Figure S2: Local intensity enhancement $|E_z|^2/|E_0|^2$ on the axis, 5 and 10 nm beyond the apex, for a plane wave polarised along the axis (BEM): rod and equal-volume sphere.

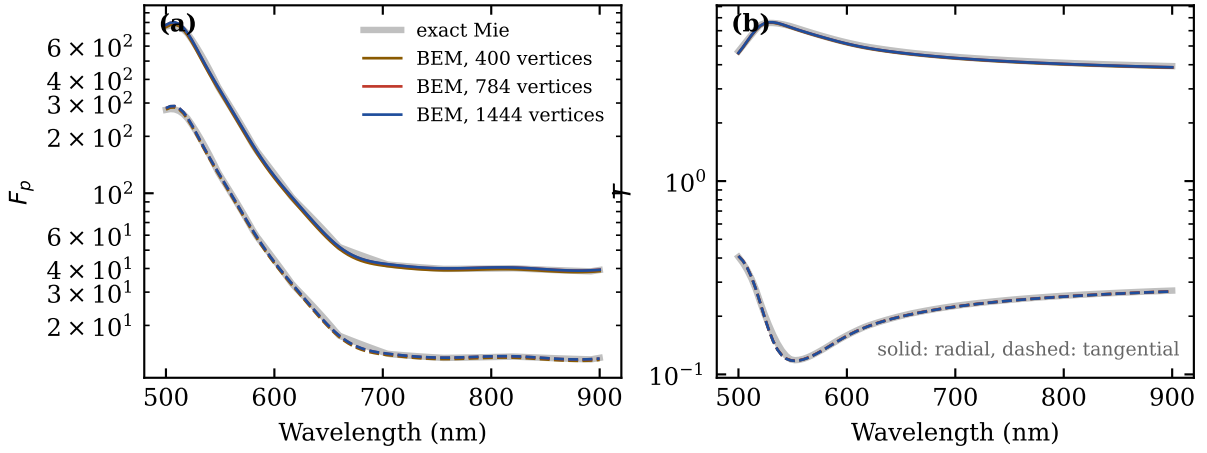


Figure S3: Validation of the BEM against the exact Mie solution for the equal-volume sphere at a 5 nm gap: (a) F_p and (b) T for the radial (solid) and tangential (dashed) dipole and three surface meshes; thick grey curves: exact results.

$\sqrt{\omega_p^2/\varepsilon_\infty - \omega(\omega + i\gamma)}/\beta$, with the tabulated Johnson–Christy ε_m , $\hbar\omega_p = 9.02$ eV, $\varepsilon_\infty = 9.84$, $\hbar\gamma = 0.071$ eV and $\beta = \sqrt{3/5} v_F$, $v_F = 1.40 \times 10^6$ m/s; integration over neighbouring elements of the two surfaces uses polar refinement (`npol` = 20). The local reference uses the same mesh. Script: `job_nonlocal.m`.

S3. Drude–Lorentz representation of Johnson–Christy gold

For the time-domain body-of-revolution solver the Johnson–Christy permittivity [1] is represented as

$$\varepsilon(f) = \varepsilon_\infty - \frac{\sigma_0 f_0^2}{f^2 + if\gamma_0} + \sum_{j=1}^2 \frac{\sigma_j f_j^2}{f_j^2 - f^2 - if\gamma_j}, \quad (\text{S1})$$

with $f = 1/\lambda$ in μm^{-1} (Meep convention, $e^{-i\omega t}$). A least-squares fit over 440 nm to 1000 nm, weighted by $|\varepsilon|^{-1/2}$, gives the parameters of Table S7; the root-mean-square relative error is 1.6 % and the maximum 3.4 %. The same model is used in the Mie reference calculations, so that benchmark differences reflect discretisation only.

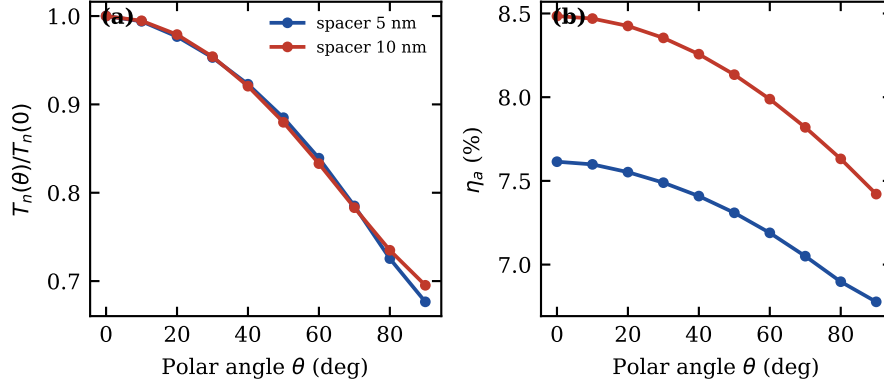


Figure S4: Emitter on a spacer shell of thickness s around the rod tip at polar angle θ , dipole normal to the surface: (a) radiative enhancement relative to the on-axis value, (b) antenna efficiency at the radiative peak (610 nm). Script: `job_offset.m`.

Table S6: Emission of the dipole next to the rod on glass ($n = 1.52$, dipole height 10.97 nm, 640 nm) by reciprocity. Radiative rate in units of the vacuum rate; for the lossless rod ($\varepsilon = 4$) it must equal the total decay rate obtained from the near field. Fractions of the radiated power.

case	gap	radiative rate	total rate	into glass	oil NA 1.45	dry NA 0.9 (top)
bare dipole x , analytic	—	1.299	1.299	82.7%	79.2%	14.7%
bare dipole x , BEM reciprocity	—	1.298	—	82.6%	78.5% ^a	14.6%
dielectric rod, x	5	2.785	2.882	82.7%	78.5% ^a	14.6%
dielectric rod, y	5	0.938	0.934	82.7%	78.5% ^a	14.6%
dielectric rod, z	5	1.516	1.421	84.9%	81.5% ^a	9.8%
gold rod, x	5	271.6	1960	82.7%	78.5% ^a	14.6%
gold rod, x	20	14.5	78.8	82.6%	78.4% ^a	14.7%

^aThe direction grid places the cone edge between quadrature points, which biases the NA-limited fractions by up to 2 percentage points; the main text quotes the analytic bare-dipole fractions, which the rod reproduces for the total fraction into the glass to < 0.2 points.

S4. Three-dimensional FDTD set-up

Lumerical FDTD 2024 R1, cubic domain of side 600 nm with 12 PML layers, maximum simulation time 300 fs, auto-shutoff 10^{-5} (reached after 19 fs to 28 fs for cases B, C, D and 52 fs to 80 fs for the axial rod cases). A uniform override mesh ($\Delta = 2$ nm unless stated) covers the particle, gap and source; global mesh accuracy 3. Radiated power is the flux through six planar monitors on a 400 nm cube; 201 wavelengths between 500 and 900 nm. The dipole sits at $z = 35$ nm, between two nodes of the 2 nm mesh, whereas in the free-space reference (case B) it sits at $z = 0$ on a node. The script that builds and runs all cases from scratch is provided with the data.

S5. Free-space reference and mesh sensitivity of the three-dimensional solver

Figure S5 shows the free-space check: T_B stays within $[0.993, 1.003]$ while $F_{p,B}$ oscillates between 0.93 and 1.13, which yields the unphysical $\eta_B = 1.08$ at 748 nm and bounds the normalisation uncertainty of F_p at $\pm 13\%$ (3% at 610 nm). Figure S6 shows the mesh sensitivity of case A: the radiative peak position (616(4) nm) and the efficiency at it are stable, whereas the peak heights depend on the sub-cell position of the source. Ratios between cases computed with the same

Table S7: Fit parameters (hcf gives energies in eV with $hc = 1.2398 \text{ eV } \mu\text{m}$).

term	$f_j \text{ (}\mu\text{m}^{-1}\text{)}$	$hcf_j \text{ (eV)}$	$\gamma_j \text{ (}\mu\text{m}^{-1}\text{)}$	σ_j
ε_∞		2.229		
Drude	8.276	10.26	0.0490	0.701
Lorentz 1	2.208	2.737	0.509	1.156
Lorentz 2	2.755	3.416	0.010	2.101

Table S8: All three-dimensional FDTD cases (raw definitions). η_{F_p} , η_T : efficiency at the Purcell and radiative peaks; $\eta_{\max}$: spectral maximum.

case	λ_{F_p}	$F_p^{\max}$	λ_T	$T_{\max}$	η_{F_p}	η_T	$\eta_{\max}$
A rod, axial, 5 nm	610	1496	616	119	7.6%	8.6%	19%
C rod, transverse	508	239	500	0.48	0.18%	0.21%	5.7%
D sphere, radial	508	659	532	6.03	0.76%	1.1%	12%
D $_\perp$ sphere, tangential	510	268	500	0.45	0.15%	0.18%	1.9%
D Mie (exact)	506	772	530	6.58	0.67%	–	11%
G3 gap 3 nm, 1-nm mesh	516	7198	614	252	0.05%	6.4%	7.7%
G10 gap 10 nm	610	553	614	46.8	8.1%	8.9%	31%
G20 gap 20 nm	610	91	618	9.9	9.6%	12%	74%
M3 mesh 3 nm	644	2027	648	184	8.7%	9.3%	18%
M1.5 mesh 1.5 nm	540	3752	620	165	0.05%	7.0%	9.6%

mesh and source position are unaffected.

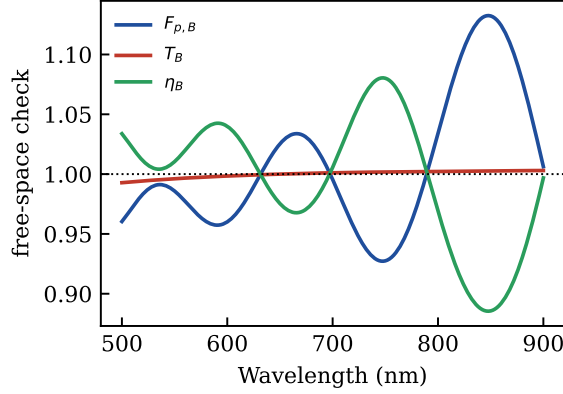


Figure S5: Free-space reference B of the three-dimensional solver.

S6. Body-of-revolution solver: method and benchmarks

Cylindrical-coordinate FDTD (Meep [2]) with field dependence $e^{im\varphi}$; axial dipole: E_z source, $m = 0$; transverse dipole: E_r source, $m = 1$. T is the ratio of the fluxes through a closed surface (top and bottom discs and a cylindrical side wall) around the particle with and without it. $F_p = (P_{\text{rad}} + P_{\text{abs}})/P_0$, where P_{abs} is the net inflow through a second closed surface that encloses only the particle, its top disc lying at mid-gap. The local density of states at the source cell gives the same order of magnitude but is spiky near the metal and is used only as a cross-check. The band 490 nm to 900 nm is assembled from three runs with pulses covering 460 to 610, 540 to 760 and 630 nm to 990 nm, of which only 490 to 570, 570 to 700 and 700 nm to 900 nm are kept: a

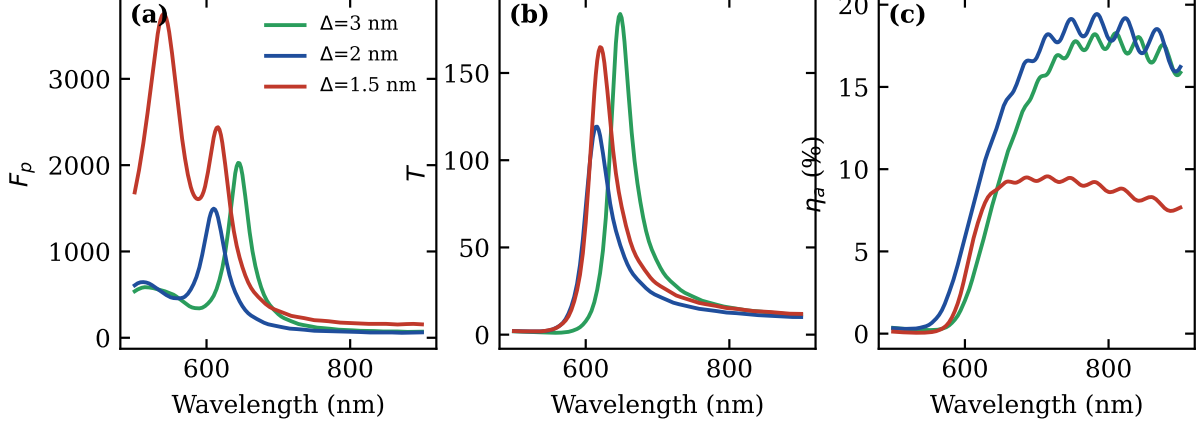


Figure S6: Mesh sensitivity of case A in the three-dimensional solver: (a) F_p , (b) T , (c) η_a for override meshes of 3, 2 and 1.5 nm.

broadband single run produced errors up to 37% at the band edges, where the source spectrum is weak. Run time 30 $\mu\text{m}/c$ after the source (identical F_p to 60); PML thickness 0.15 μm , with a 0.3 μm check at the finest grid. To avoid long-lived numerical ringing, the Lorentz-pole damping in the gold fit is bounded below by 0.2 μm^{-1} (same fit accuracy).

Benchmark against the exact Mie solution for a dielectric sphere ($\varepsilon = 4$, $R = 15.874$ nm, radial dipole at 5 nm): a grid of 2 nm reproduces F_p within 1% in the band centres; a 4 nm grid gives a uniform 4% to 5% overestimate with a maximum deviation of 12%. For gold (radial dipole, 5 nm gap) at a 2 nm grid, T deviates from Mie by about 10% while the power-balance F_p is about three times too large; at a 20 nm gap the F_p error drops to 24% and T agrees within 5%. The F_p error is therefore a near-field discretisation error of the staircased metal surface. Grids of 2, 1.33 and 1 nm with linear extrapolation to zero spacing are summarised in Table S9.

Table S9: Body-of-revolution solver against exact Mie theory (gold sphere, radial dipole, 5 nm gap) and grid convergence of the nanorod (axial dipole, 5 nm gap). Errors are medians over 490 nm to 900 nm.

grid (nm)	sphere: $ \Delta F_p $	sphere: $ \Delta T $	rod: $T_{\max} (\lambda_T)$	rod: NA 0.9 fraction
2.00	215%	10.2%	149 (631 nm)	0.181
1.33	108%	8.4%	134 (617 nm)	0.181
1.00	93%	7.2%	125 (611 nm)	0.180
extrapolated	48%	4.4%	—	—

PML thickness 0.15 versus 0.3 μm at the 1 nm grid changes T by 0.1% (median) but F_p by 9%. The radiative quantities are therefore converged to the level quoted in the main text, whereas F_p in a 5 nm gap is not converged at grids of 1 nm with a staircased surface and is not used; all F_p values in the main text are BEM results. The transverse ($m = \pm 1$) configuration with a single E_r source on the axis gave unphysical (negative) total rates and is likewise not used; all transverse results in the main text are BEM results. All runs were executed on GitHub-hosted runners (workflow [.github/workflows/meep-runs.yml](https://github.com/workflows/meep-runs.yml)); raw outputs are in `isi/results/ci`.

References

- [1] P. B. Johnson and R. W. Christy, Phys. Rev. B **6**, 4370 (1972).
- [2] A. F. Oskooi *et al.*, Comput. Phys. Commun. **181**, 687 (2010).

- [3] U. Hohenester and A. Trügler, Comput. Phys. Commun. **183**, 370 (2012).
- [4] J. Waxenegger, A. Trügler and U. Hohenester, Comput. Phys. Commun. **193**, 138 (2015).
- [5] U. Hohenester, Comput. Phys. Commun. **222**, 209 (2018).
- [6] R. Ruppin, J. Chem. Phys. **76**, 1681 (1982).
- [7] Y. S. Kim, P. T. Leung and T. F. George, Surf. Sci. **195**, 1 (1988).
- [8] Y. Luo *et al.*, Phys. Rev. Lett. **111**, 093901 (2013).